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2.Euclid unit

PDF Documents

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Euclid's Division Lemma

Euclid's division lemma states that for any two positive integers, say 'a' and 'b', the condition ' $a = bq + r$ ', where $0 \leq r < b$ always holds true. Mathematically, we can express this as 'Dividend = (Divisor $\times$ Quotient) + Remainder'. A lemma is a statement that is already proved. Euclid, a Greek mathematician, devised Euclid's division lemma.

Euclid's Division Lemma

Euclid's Division Lemma (lemma is similar to a theorem) says that, for given two positive [integers](#), 'a' and 'b', there exist unique integers, 'q' and 'r', such that: $a = bq + r$, where $0 \leq r < b$.

Euclid's Division Lemma



$$a = bq + r, 0 \leq r < b$$

$$\begin{array}{ccccccc} & \text{Divisor}(b) & & \text{Remainder}(r) & & & \\ & \downarrow & & \downarrow & & & \\ 48 & = & 5 & \times & 9 & + & 3 \\ \uparrow & & & & \uparrow & & \\ \text{Dividend}(a) & & & & \text{Quotient}(q) & & \end{array}$$

The integer 'q' is the [quotient](#) and the integer 'r' is the [remainder](#). The quotient and the remainder are **unique**. In simple words, Euclid's division lemma statement is that if we divide an integer by another non-zero integer, we will get a unique integer as quotient and a unique integer as remainder.

We can write the above scenario mathematically as: '**Dividend = (Divisor $\times$ Quotient) + Remainder**'. The above scenario shows the way of representing the division of positive integers with the help of Euclid's division lemma.

Here are some examples of the application of Euclid's division lemma:

$$23 = (2 \times 11) + 1$$

$$54 = (7 \times 7) + 5$$

$$63 = (9 \times 7) + 0$$

Euclid's Division Lemma Proof

Euclid's lemma or Euclid's division lemma statement says that for given two positive integers, 'a' and 'b', there exists unique integers 'q' and 'r' such that, $a = bq + r$, $0 \leq r < b$.

So, let's take $a = 9$ and $b = 1$.

$$9 = (1 \times 9) + 0$$

Here, $q = 9$ and $r = 0$, and we can clearly see that $0 \leq r < 1$.

Now, let's take $a = 9$ and $b = 2$.

$$9 = (2 \times 4) + 1$$

Here, $q = 4$ and $r = 1$, and we can clearly see that $0 \leq r < 2$.

Now, let's take $a = 9$ and $b = 3$.

$$9 = (3 \times 3) + 0$$

Here, $q = 3$ and $r = 0$, and we can clearly see that $0 \leq r < 3$.

Now, let's take $a = 9$ and $b = 4$.

$$9 = (4 \times 2) + 1$$

Here, $q = 2$ and $r = 1$, and we can clearly see that $0 \leq r < 4$.

We can observe that when we divide two integers using the Euclid division lemma, we always get a non-negative integral remainder less than the divisor.

How to Find HCF By Euclid's Division Lemma?

Euclid's division lemma has many uses or applications. Let's learn about two of the most important uses of this algorithm. We use Euclid's division lemma to find the [HCF](#) of large numbers which is typically difficult to calculate using basic HCF calculation techniques. The HCF of two numbers can be calculated with the help of Euclid's division lemma by following these steps.

Let's take two numbers ' c ' and ' d ' for which we need to find the HCF such that $c > d$.

- **Step 1:-** Apply Euclid's division lemma to ' c ' and ' d '. We can find [whole numbers](#), ' q ' and ' r ' such that $c = dq + r$, $0 \leq r < d$.
- **Step 2:-** If $r = 0$, ' d ' is the HCF of ' c ' and ' d '. If $r \neq 0$, apply the division lemma again to ' d ' and ' r '.
- **Step 3:-** Continue the process till the remainder is zero. The divisor at this stage will be the required HCF.

Let us use Euclid's division lemma to find the HCF of 42 and 64. The solution is shown in the figure below.

HCF Using Euclid's Division Lemma



Euclid's Division Lemma is $c = dq + r, 0 \leq r < d$

HCF of 42 and 64 using Euclid's Division Lemma is:

Since $64 > 42$, $c = 64$ and $d = 42$

According to Euclid's Division Lemma, $c = dq + r, 0 \leq r < d$

Substituting the values of 'c' and 'd', we get.

$$\begin{array}{l} 64 = (42 \times 1) + 22 \\ 42 = (22 \times 1) + 20 \\ 22 = (20 \times 1) + 2 \leftarrow \text{HCF of (42, 64)} \\ 20 = (2 \times 10) + 0 \leftarrow \text{Stop here} \end{array}$$

Since the remainder is now zero, we can stop the process. The HCF of 42 and 64 is 2. When we apply Euclid's division lemma repeatedly to find the HCF, this process is known as [Euclid's division algorithm](#).

Euclid's Division Lemma and Properties of Numbers

We can understand the various properties of odd and [even numbers](#) using Euclid's division lemma. For example, let us show that every positive even integer is of the form $2q$ and that every positive odd integer is of the form $2q + 1$, where q is some integer. Assume a is a positive integer and $b = 2$.

By Euclid's division lemma, we can write:

$$a = 2q + r$$

where $q \geq 0$ and r can take either 0 or 1 by the rule $0 \leq r < b$.

Thus, $a = 2q$ or $a = 2q + 1$.

If 'a' is of the form ' $2q$ ', then 'a' is an even integer, and if 'a' is of the form $2q + 1$, then it is an odd integer.

For example, if we substitute $q = 1, 2$, and 3 , then $a = 2q$ is even.

$$a = 2 \times 1 = 2$$

$$a = 2 \times 2 = 4$$

$$a = 2 \times 3 = 6$$

And, if we substitute $q = 1, 2,$ and $3,$ we will find that $a = 2q + 1$ is odd.

$$a = (2 \times 1) + 1 = 3$$

$$a = (2 \times 2) + 1 = 5$$

$$a = (2 \times 3) + 1 = 7$$